

P3↔P2 Structural Co-Adaptation Inheritance

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2026

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Attribution

This work derives from and formalizes commitments across the CKS theory trilogy: The Instinct/Reasoning Separation Outside the Model (Li, April 2026), Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026), and Inter-Self Coordination via Shared Substrate / Full Aspect Integration (Li, April 2026).

Abstract

Paper 2 commits to structural co-adaptation (C1.30): a CKS-governed Self can evolve its governance architecture — creating, merging, splitting, or dissolving aspects; reorganizing cell membership; adjusting Self-level integration — as operational needs change, with all structural changes governed through lifecycle primitives. This note formalizes the inheritance edge by which Paper 3 extends that commitment to include inter-Self learning sources. Paper 3's Full Aspect Integration (FAI) mechanism produces conflict annotations and action-feedback records that reveal governance boundary issues invisible from intra-Self experience alone. These outputs feed Paper 2's existing structural decision process as new information inputs. The structural change mechanisms themselves — lifecycle events, substrate-edit operations, directed selection under home governance — are unchanged from Paper 2. What Paper 3 adds is the information that informs structural decisions: conflict patterns from inter-Self coordination and DNA patterns encountered during aspect exchange. The extension follows the "same mechanisms, new sources" pattern that Series CC notes CC.10 and CC.14 established for evolution mechanisms and vertical evolution respectively; CC.15 is a fourth instance of that pattern applied to structural co-adaptation.

1. Inheritance identification

Source commitment. Paper 2 C1.30 establishes structural co-adaptation as an architectural property of CKS-governed Selves: the governance architecture supports structural evolution as operational needs change. Aspects can be created, merged, split, or dissolved as coordination domains evolve. Cell membership in aspects can be reorganized. Self-level integration architecture can evolve. These changes are governed through Paper 2's lifecycle primitives — the same birth, mating, and death operations that apply to cells apply at aspect and Self scope — and through Paper 2's vertical evolution axis, where reorganization operates as a substrate-edit operation under the authority

architecture Paper 2's Claims 4 and 5 establish. Structural co-adaptation at intra-Self scope means an organization's AI governance architecture can remain coherent with its operational context as that context changes over time.

Extension. Paper 3 extends structural co-adaptation by providing inter-Self learning inputs — specifically, outputs from FAI events — that inform structural decisions within each participating Self's home governance. Paper 3's D2.55 covers FAI-informed aspect restructuring: conflict carry-through annotations and action-feedback records from FAI events reveal governance boundary misalignments in a Self's current aspect structure, informing which aspects to create, merge, or split. Paper 3's D2.56 covers FAI-informed Self architecture evolution: conflict patterns accumulating at the Self scope may reveal integration architecture issues, informing directed selection at the Self level — potentially adjusting cross-aspect coordination rules, refining the Self's purpose statement, or modifying expression mechanism parameters. Paper 3 also formalizes the structural co-adaptation cycle: the pathway from FAI events through evolution feed through home governance proposals through structural decisions and back to improved future FAI contributions.

2. Source commitment at Paper 2 scope

Paper 2 C1.30 locates structural co-adaptation within Claim 6's enterprise brain vision and Claim 4's evolution architecture. The load-bearing architectural property is that structural reorganization is available as a substrate-edit primitive. Because the orchestration substrates that define aspect composition are themselves substrate content governed under Paper 2's authority architecture, reorganizing structure is not categorically different from evolving content. Both proceed through governed substrate modification; both are available on operational timescales; both are subject to the multi-shaped governance Claim 5 establishes.

At intra-Self scope, the information that drives structural co-adaptation is intra-Self operational experience. Aspects develop friction — cells whose responsibilities have drifted from the aspect's coordination domain, aspects whose boundaries no longer match how coordination actually flows, integration patterns that have become overhead. Action-feedback evolution records this friction as lived experience; governance proposals translate it into structural decisions; directed selection enacts the decisions through substrate-edit operations. The cycle operates entirely within the home governance perimeter, with no visibility into how other Selves have structured comparable coordination domains. That intra-Self scope is the exact boundary Paper 3 extends — not the mechanisms, but the information space available to them.

3. Extension to inter-Self scope

Paper 3 extends structural co-adaptation by opening two new information channels through which FAI events inform structural decisions.

Channel 1: Conflict carry-through as governance boundary signal. When a Self participates in an FAI event, the three-tier conflict-handling mechanism may produce conflict carry-through annotations — conflicts surfaced during the event that no orchestration rule resolved and that were preserved within the home substrate as first-class state. These annotations are not merely records of inter-Self friction; they are evidence about the home Self's current aspect structure. A conflict that recurs across multiple FAI events at a particular coordination boundary is a signal that the boundary may be misdrawn in the home Self's aspect structure — that the aspect covering that domain may need to be split, merged with an adjacent aspect, or reconstituted with different cell membership. Before Paper 3, this class of signal did not exist because there were no inter-Self events to generate it.

Channel 2: DNA absorption patterns as structural gap detector. FAI events give each participating Self access to orchestration patterns, schemas, and rules from other Selves' contributed aspects. When a home Self's governance authorizes selective absorption of DNA content from the shared substrate, the governance process must decide where in the home Self's aspect structure the absorbed content belongs. This absorption decision is itself structural information. DNA content from another Self that does not fit cleanly into any existing home aspect — that falls between aspects or across multiple aspects' claimed domains — is evidence of a structural gap: a coordination domain the home Self's current aspect structure does not cleanly bound. FAI-informed aspect restructuring (D2.55) uses this evidence to inform decisions about aspect creation, merger, or boundary adjustment.

The structural co-adaptation cycle. Paper 3 formalizes an extended co-adaptation cycle that makes the inter-Self learning loop explicit:

FAI events → evolution feed → home action-feedback ingestion (including conflict annotations) → governance proposals for structural improvement → directed selection (which may include aspect restructuring) → improved aspect structure → better-aligned future FAI contributions

This cycle extends Paper 2's intra-Self adaptation cycle by inserting the FAI ring at the beginning. The extended cycle's right half — governance proposals → directed selection → structural improvement — is Paper 2's cycle unchanged. The extension is the left half: FAI events now feed the action-feedback and DNA evolution inputs that reach the governance proposal stage. The cycle notation makes the extension explicit and traceable: one can identify exactly where inter-Self learning enters the existing intra-Self cycle, and one can verify that it enters through Paper 2's existing evolution mechanisms rather than through a new mechanism.

4. What is preserved and what is extended

Preserved. Every structural change mechanism is a Paper 2 mechanism operating as Paper 2 specifies. Aspects are created, modified, and dissolved through governed lifecycle events. Cell membership in aspects is reorganized through substrate-edit operations under

the authority architecture Paper 2's Claims 4 and 5 establish. Self-level integration architecture evolves through vertical evolution's substrate-edit primitives. The decision authority for all structural changes remains with home governance. No FAI event can directly impose a structural change on a participating Self; FAI outputs enter the home governance process as information, and home governance decides what structural changes, if any, follow.

The authority-vs-labor distinction Paper 1 §3.3 establishes holds throughout: humans hold authority over structural decisions; the labor of analyzing FAI outputs and drafting structural proposals is available to humans directly or to the substrate mediator operating under human direction.

Extended. The information available to inform structural decisions is expanded. Before Paper 3, structural co-adaptation was informed only by intra-Self operational experience — friction visible within the home governance perimeter. After Paper 3, it is also informed by inter-Self coordination experience: conflict patterns that crossed the inter-Self boundary and were annotated in the home substrate, and DNA absorption decisions that revealed structural gaps in the home Self's aspect coverage. The structural governance mechanisms are unchanged; the information reaching them is richer.

5. Prior-art inheritance claim

Paper 3's FAI-informed structural co-adaptation is an extension of Paper 2's structural co-adaptation commitment (C1.30), not a new architectural mechanism. The extension adds inter-Self learning sources to the information space that informs structural decisions within each home governance perimeter. Every structural change mechanism Paper 3's co-adaptation touches — lifecycle primitives, substrate-edit operations, directed selection, multi-shaped governance over structural decisions — is a Paper 2 mechanism. Paper 3's contribution is the FAI ring that supplies conflict annotations and DNA absorption patterns as new inputs to those mechanisms.

This note is the fourth instance of the "same mechanisms, new sources" inheritance pattern in Series CC. CC.10 established that Paper 3 extends Paper 2's evolution mechanisms by adding a FAI-boundary input locus rather than defining new mechanisms. CC.14 established that Paper 3 extends Paper 2's vertical evolution by adding inter-Self DNA exchange as a new source of substrate content to reorganize, while vertical evolution's substrate-edit primitives remain unchanged. CC.15 establishes that Paper 3 extends Paper 2's structural co-adaptation by adding FAI-produced conflict annotations and DNA absorption patterns as new information sources for structural decisions, while the structural change mechanisms remain unchanged.

The consistency of this pattern across four CC.1x notes strengthens the inheritance argument. Paper 3 consistently extends Paper 2 by providing new information sources at the inter-Self boundary that feed existing intra-Self mechanisms, never by defining new mechanisms requiring independent architectural defense. The prior-art consequence is

correspondingly consistent: Paper 3's inter-Self learning inputs to structural co-adaptation are novel, but the structural co-adaptation architecture they feed is Paper 2's. Claims of novelty attach to the FAI outputs and their routing to home governance; the co-adaptation mechanisms they reach were established at Paper 2.

Source papers

Li, W. (2026a). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026b). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026c). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *P3↔P2 Structural Co-Adaptation Inheritance*. May 15, 2026. ORCID: 0009-0004-8065-3235.